

Radar based Contactless Blood Pressure Estimation System using Signal Decomposition and Deep Neural Network

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Abstract—Non-contact blood pressure monitoring holds significant importance in terms of convenience, real-time monitoring, and long-term surveillance, and it plays a positive role in fields of blood pressure management and disease prevention. This paper proposes a blood pressure monitoring method based on a novel signal processing method with frequency-modulated continuous wave (FMCW) radar. Firstly, we employ the millimeter-wave radar to collect the phase signal of the neck. Then, we present an empirical mode decomposition method based on time-varying filters to decompose the phase signal, and propose to select efficient Intrinsic Mode Functions (IMFs) for blood pulse-wave signal reconstruction. Finally, this paper proposes a dual-stream fusion neural network for pulse-wave features extraction and blood pressure estimation. The validation of the proposed method's feasibility and effectiveness underscores its comparative advantage over the most advanced techniques in the field, as evidenced by the results.

Index Terms—Frequency-modulated continuous wave, contactless, blood pressure monitoring, signal decomposition, deep learning.

I. INTRODUCTION

WITH the development of radar technology, radar has been widely applied in industrial environments for detecting personnel vital signs, enabling remote health monitoring of workers as well as safety and emergency response. Monitoring of blood pressure is crucial for maintaining cardiovascular health and preventing chronic diseases. According to the statistics from the World Health Organization (WHO), high blood pressure is one of the leading risk factors globally for heart disease and stroke [1]. The American Heart Association (AHA) emphasizes that elevated blood pressure serves as a

significant contributor to various cardiovascular health conditions, notably coronary heart disease, cardiac insufficiency, and renal disorders [2].

For routine clinical blood pressure monitoring, non-invasive methods using cuff-based devices remain the primary means of measurement [3]. Non-invasive blood pressure monitoring methods usually includes traditional manual methods and modern automatic electronic based blood pressure monitors. Manual and automatic electronic devices are widely accepted in daily medical practice, offering high accuracy while being more convenient and safe. However, the aforementioned methods are based on the cuff-based contact measurement, and these methods are not suitable for special populations such as burn victims, obese patients, and infants. The radar-based contactless blood pressure monitoring system does not require direct contact with patients, making it suitable for the detection of sensitive populations and infectious patients. It is not restricted by cuff size, wearing position, or ambient light conditions, and is easy to operate, supporting long-term dynamic monitoring.

Radar has been proven to be feasible for the monitoring of vital signs [4]–[7]. Millimeter-wave radar can detect subtle skin motion for monitoring physiological signals such as respiration and heartbeat. In recent years, non-contact blood pressure monitoring based on radar has attracted attention. The author in [8] employs a Doppler radar to acquire the heartbeat signal and calculates the systolic blood pressure (SBP) and diastolic blood pressure (DBP) based on the periods of heartbeat and its contractions. However, the radar used has a sampling frequency of only 100 Hz, which may not promptly capture the changes in the waveform of the blood. In order to investigate the relationship between pulse-waves and blood pressure, certain research endeavors propose a potential relationship existing between Pulse Transit Time (PTT) and blood pressure [9]. In [10], the authors extract the surface micro-displacement caused by the central artery of the aorta using a Doppler radar. Subsequently, the PTT is extracted to calculate SBP and DBP. In [11], the authors present a non-contact continuous blood pressure monitoring method by using the PTT obtained from both the neck and chest. In [12], the authors utilizes a wearable 120 GHz Doppler radar sensor to collect the pulse signals of the human elbow, enabling the extraction of the Reflection Pulse Transit Time (RPTT) for blood pressure monitoring. Moreover, the variability in PTT time due to individual differences results in the blood pressure

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measurement method based on PTT having poorer robustness.

Recently, researchers have turned their attention to utilize machine learning to address the blood pressure monitoring issue. Machine learning can automatically extract useful features from the pulse-wave for blood pressure prediction, and they possess robustness across different individuals. Sannino et al [13] utilize genetic programming to infer the interconnections between blood pressure values and pulse-wave. Lee et al [14] propose an innovative deep learning model that integrates Deep Belief Network (DBN) with Deep Neural Network (DNN), aimed at comprehensively understanding and learning the complex and nonlinear relationship between the features extracted from pulse-wave signals and human blood pressure. The literature [15] utilizes a 60 GHz radar to extract the wrist pulse pressure waves, capturing the wrist feature information along with calibrated parameters for human age, gender, height and weight. These parameters are collectively used as inputs of a designed neural network-based regression model blood pressure prediction. Ran et al [16] employ a millimeter-wave radar to collect the human wrist pulse-waves, and design an encoder-decoder neural network to extract the blood pressure.

Despite the optimistic outlook of these methods, there are still some issues that need to be resolved. On the one hand, the phase signal usually contains a significant amount of interference and noise, the pulse-wave is highly sensitive to noise, and it is challenging for neural network methods to accurately predict blood pressure. The authors in [17] conduct filtering and smoothing operation on the radar signals, and automatically extract the characteristic parameters of arterial pulse-waves. The authors carry out correlation analysis to further explore the interconnections between these parameters and blood pressure. However, the filtering and smoothing operation cannot eliminate the influence of interference and noise. Previous literature has used wavelet transform to denoise pulse-wave signals. However, how to select wavelets for the reconstruction of pulse-waves after decomposition is a question worth considering. **If low-quality pulse wave signals are input into the neural network, the uncertainty in the learning process increases.** The network will not only require significant computational resources and training time to learn how to distinguish noise from pulse wave signals, but also fail to guarantee detection performance. Therefore, a contactless blood pressure monitoring framework should fully consider the impact of interference and noise. On the other hand, designing a deep learning model to extract useful feature information from pulse-waves is another challenge. In [18] and [19], the authors adopt multiple convolutional kernels of varying sizes to extract multi-scale characteristics, achieving good results. However, fixed-size convolutional kernels can only extract features within their fixed receptive fields, lacking flexibility. When faced with data of different sizes, the effectiveness of the mentioned method may be impacted. More importantly, the current radar-based blood pressure monitoring mostly uses single-pulse pulse-wave signals, without considering the long-term dependency between continuous pulse-waves. This observation inspired us to address two critical issues. The first one is how to design a more flexible and adaptable feature extraction mechanism that can overcome the limitations of

fixed-size convolutional kernels and dynamically capture the multi-scale features across diverse pulse-wave data. The other one is how to incorporate the long-term dependencies in the continuous pulse-wave signals into the radar-based blood pressure monitoring framework.

Therefore, this paper proposes a framework for accurate monitoring of human blood pressure based on Frequency-Modulated Continuous Wave (FMCW) radar. The main contributions of this paper are as follows:

- With the extracted phase signal of the human body, we decompose the phase signal into several Intrinsic Mode Functions (IMFs) components by using the time varying filter based empirical mode decomposition (TVF-EMD) [20]. Then, we propose to select three key IMFs for blood pulse-wave signal construction.
- We design a novel dual-stream fusion neural network based blood pressure estimation (DSFNN-BP) method to extract pulse-wave signal feature for blood pressure prediction. The S-Stream is designed based on the spatial attention mechanism to extract shallow features, while the L-Stream is based on the convolutional module to learn advanced features. The features extracted from the dual streams are further inputted into the introduced multi-head attention mechanism module to extract different time scale features for blood pressure prediction.
- We design experiments to substantiate the effectiveness of the proposed method, yielding mean absolute errors (MAE) of 4.16 mmHg for SBP and 3.87 mmHg for DBP, both adhering to the standard set by the Association for the Advancement of Medical Instrumentation (AAMI). **Moreover, experiments conducted at distances and different angles have demonstrated the robustness of our proposed method.**

The remaining parts of this paper are organized as follows. Section II introduces the proposed blood pressure prediction system. Section III validates the effectiveness of the method, and concluding remarks are provided in Section IV.

II. PROPOSED BLOOD PRESSURE MONITORING METHOD

A. System Overview

The proposed blood pressure estimation framework contains three steps.

Step 1: Human neck positioning and phase extraction: The FFT is performed on the intermediate frequency (IF) signal to obtain the range-time map, and then the phase signal at the human neck is extracted.

Step 2: Interference and noise suppression: The phase signal is decomposed into multiple IMFs using TVF-EMD, and the potential pulse-wave IMFs are efficiently selected based on IMFs selection criteria for reconstructing the pulse-wave signal.

Step 3: Blood pressure estimation: The pulse-wave sequence is inputted into the designed dual-stream fusion neural network to extract shallow and deep features of the pulse-wave, and multi-head attention is used to further comprehend the complex dependencies among pulse-wave sequences for blood pressure estimation.

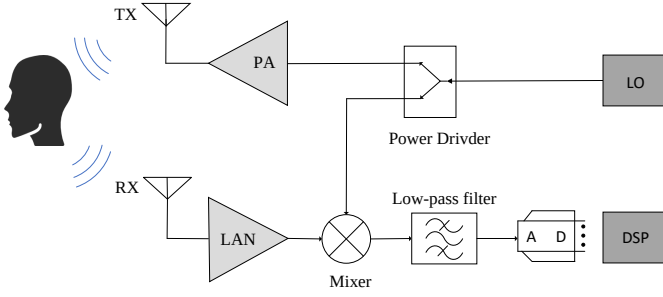


Fig. 1: Schematic of the FMCW radar system

B. FMCW Radar Signal Model

This paper utilizes a linearly frequency-modulated continuous wave as the transmitted signal to illuminate the human pulse-wave. The typical one-transmitter-one-receiver antenna system model is shown in Fig. 1. The reflected echo signals from the human neck are then received by the receiver. After modulating to produce the mixed signal by a mixer, the high-frequency components are filtered out by the low-pass filter to obtain an IF signal. Finally, the IF signal is sampled by an analog-to-digital converter (ADC) and then sent to the signal processing module for further processing.

The transmitted signal can be expressed as:

$$X_T(t) = A_T \exp(j2\pi(f_c t + \frac{B}{2T}t^2) + \theta(t)), \quad (1)$$

where A_T represents the transmitted signal's amplitude, f_c is the starting frequency of the transmitted signal, B is the signal bandwidth, T is the temporal span of a single chirp, and $\theta(t)$ is the initial phase.

The distance between the human neck and the radar antenna is d_0 , the pulsatile motion is $x(t)$, and the speed of the light is c . Then, the propagation time t_d of the signal from transmission to reception is given by:

$$t_d = \frac{2(d_0 + x(t))}{c}. \quad (2)$$

The mathematical expression for the received signal is:

$$X_R(t) = A_R \exp(j2\pi(f_c(t - t_d) + \frac{B}{2T}(t - t_d)^2) + \theta(t - t_d)). \quad (3)$$

Then, the transmitted signal and the received signal are mixed and filtered to produce the IF signal $S_{IF}(t)$:

$$\begin{aligned} S_{IF}(t) &= A_T A_R \exp(j(2\pi\frac{B}{T}t_d t + 2\pi f_c t_d + \pi\frac{B}{T}t_d^2 + \Delta\theta(t))) \\ &\approx A_T A_R \exp(j(2\pi\frac{B}{T}t_d t + 2\pi f_c t_d + \Delta\theta(t))) \\ &\approx A_T A_R \exp(j(2\pi\frac{2BR}{cT}t + \frac{4\pi(x(t)+d_0)}{\lambda})) \\ &= A_T A_R \exp(j(2\pi f_b t + \psi(t))), \end{aligned} \quad (4)$$

where $\Delta\theta(t)$ is noise residue item, f_b is the frequency of intermediate frequency signals.

The first approximation in (4) is due to the fact that t_d is generally less than 1 ns, $\frac{B}{T}$ is the slope of signals and $\pi\frac{B}{T}t_d^2$ can be neglected. For the second approximation, $\Delta\theta(t)$ can be neglected due to the distance-related effects [21], and the amplitude of pulsatile motion is extremely weak, so $x(t)t$ can

also be neglected. Therefore, the phase signal in the IF signal can be expressed as:

$$\psi(t) = 4\pi \frac{x(t)}{\lambda}. \quad (5)$$

Despite the extremely minute skin displacement induced by the pulse, we are able to observe this displacement through measuring the displacement in the phase of the IF signal. This paper tries to apply this displacement change for blood pressure monitoring.

C. Human Neck Localization and Phase-wave Signal Extraction

In this subsection, we preprocess the raw phase signals to extract the phase signal of the blood pulse.

(i) Human neck localization: The first step of finding the pulse-wave signal is identifying the radar echoes from the human neck. Radar data is stored in a two-dimensional matrix, where each chirp is sampled and stored in the rows of the matrix, and data of multiple chirps in a frame are stored in the columns. By performing FFT on each row of the matrix, we can obtain the range information of the human neck in the current chirp. Once the range bin of each chirp is determined, we extract the raw phase signal of the neck.

(ii) Pulse-wave signal extraction: Before extracting the phase information of pulse-wave, it is necessary to eliminate the non-linear distortions. The impact of radar circuit defects and static object interference can introduce the Direct Current (DC) offset, which can lead to signal distortion [22]. A high-quality, undistorted pulse-wave signal is a prerequisite for blood pressure monitoring, and it is necessary to remove the DC component in the phase signal of pulse-wave. In this paper, we apply the circular dynamic tracking algorithm [23] to optimize the fitting results of the circle by using the least squares and gradient descent methods. Once the center and radius are determined, the DC offset are eliminated.

To prevent phase ambiguity and phase wrapping issues, we apply the extended differentiate and cross-multiply (DACM) algorithm [24]. The DACM algorithm discretely implements the signal inverse tangent differentiation followed by integration:

$$\varphi[k] = \sum_{k=2}^n \frac{I[k]\{Q[k] - Q[k-1]\} - \{I[k] - I[k-1]\}Q[k]}{I[k]^2 + Q[k]^2}. \quad (6)$$

where $I[k]$ and $Q[k]$ are the I and Q signals of the k -th frame, respectively, and n is the number of frames. The phase signal $\varphi[k]$ is then embedded in $X(t)$.

It is noted that by calculating the blood pressure using the phase signal is a challenging task. This is because the details features of the original pulse-wave is usually masked by the introduced interference and noise in the environment. In previous works, this aspect has largely been overlooked. Lack of noise training may lead to poor performance of the model when facing small variations or disturbances in the input, reducing the robustness of the model. A complete and clear pulse-wave is a prerequisite for accurately blood

pressure monitoring. Therefore, $X(t)$ needs further processing to recover the original pulse-wave from the phase signal.

D. TVF-EMD based Pulse-Wave Signal Reconstruction

1) TVF-EMD based phase signal decomposition: Although deep neural networks have powerful nonlinear modeling capabilities and can directly process noisy data, the key features of the pulse-waves can be easily masked by the interference and noise, resulting in unsatisfactory accuracy under noisy conditions. Previous works in vital signs detection, it is commonly to extract the target signal from the raw signal using mode decomposition methods [25]–[27]. These mode decomposition methods decompose the signals into several IMFs components with different scaled characteristics. Respiratory and heartbeat signals are often regarded as sinusoidal periodic signals, whereas the pulse-wave signals are non-stationary. The EMD method usually encounters mode mixing issues, and the pulse-wave cannot be separated from noise. Although CEEMDAN partially addresses the mode mixing problem, it still cannot fundamentally eliminate the phenomenon of mode mixing phenomenon due to inherent limitations in the recursive decomposition principle. Considering the characteristics of pulse-wave signals, we propose to apply TVF-EMD algorithm to enhance the performance in handling the non-stationary pulse-wave signals. The TVF-EMD algorithm exhibits strong noise robustness and consistently maintains time-varying characteristics, providing unique advantages in handling the non-stationary vibration signals. The fundamental concept of the algorithm involves identifying the local cutoff frequencies of the signal and then applying time-varying filtering. The purpose of separating the local cutoff frequencies in the frequency domain is to establish time-varying filters. The signal is decomposed into local high-frequency and low-frequency components by filtering process. Through continuous time-varying filtering, IMF components with local narrowband characteristics are extracted. The detailed steps of TVF-EMD algorithm are as follows:

Step 1: We compute the instantaneous amplitude $A(t)$ and instantaneous frequency $\varphi'(t)$ of the phase signal $X(t)$:

$$A(t) = \sqrt{X(t)^2 + \hat{X}(t)^2}, \quad (7)$$

$$\varphi'(t) = d(\arctan[\hat{X}(t)/X(t)]/dt, \quad (8)$$

where $\hat{X}(t)$ is the Hilbert transform of $X(t)$.

Step 2: We find the local maxima $A(\{t_{\max}\})$ and local minima $A(\{t_{\min}\})$ of $A(t)$.

Step 3: We carry out local interpolation on $A(\{t_{\min}\})$ and $A(\{t_{\max}\})$ to obtain $\beta_1(t)$ and $\beta_2(t)$. Then, we calculate the instantaneous mean $\alpha_1(t)$ and instantaneous envelope $\alpha_2(t)$ as:

$$\alpha_1(t) = \frac{\beta_1(t) + \beta_2(t)}{2}, \quad (9)$$

$$\alpha_2(t) = \frac{\beta_2(t) - \beta_1(t)}{2}. \quad (10)$$

Step 4: We compute the interpolation results of $A^2(\{t_{\min}\})\varphi'(\{t_{\min}\})$ and $A^2(\{t_{\max}\})\varphi'(\{t_{\max}\})$, and denote by $\eta_1(t)$ and $\eta_2(t)$, respectively. Then, we compute the instantaneous frequency components $\varphi'_1(t)$ and $\varphi'_2(t)$ as:

$$\varphi'_1(t) = \frac{\eta_1(t)}{2\alpha_1^2(t) - 2\alpha_1(t)\alpha_2(t)} + \frac{\eta_2(t)}{2\alpha_1^2(t) + 2\alpha_1(t)\alpha_2(t)}, \quad (11)$$

$$\varphi'_2(t) = \frac{\eta_1(t)}{2\alpha_2^2(t) - 2\alpha_1(t)\alpha_2(t)} + \frac{\eta_2(t)}{2\alpha_2^2(t) + 2\alpha_1(t)\alpha_2(t)}. \quad (12)$$

Step 5: We compute the local cutoff frequency as:

$$\varphi'_{bis} = \frac{\varphi'_1(t) + \varphi'_2(t)}{2} = \frac{\eta_2(t) - \eta_1(t)}{4\alpha_1(t) - \alpha_2(t)}. \quad (13)$$

Step 6: Since the obtained φ'_{bis} from step 5 may be influenced by intermittent noise, we adjust the local cutoff frequency for recalibrating φ'_{bis} .

Step 7: We use the extremal points of $h(t)$ to construct B-spline interpolation:

$$h(t) = \cos\left[\int \varphi'_{bis}(t)dt\right]. \quad (14)$$

We carry out time-varying filtering on $X(t)$ to obtain the local mean function $m(t)$.

Step 8: We compute the instantaneous narrowband signal $\theta(t)$. When the stopping criterion is satisfied, we obtain multiple IMFs. It is noted that the stopping criterion $\theta(t)$ represents an instantaneous narrowband signal:

$$\theta(t) = \frac{B_{Loughlin}(t)}{\varphi_{avg}(t)}, \quad (15)$$

where $B_{Loughlin}(t)$ and $\varphi_{avg}(t)$ are, respectively, the Loughlin instantaneous bandwidth and the weighted average of the instantaneous frequency of each component signal, and they are defined as:

$$\varphi_{avg}(t) = \frac{\alpha_1^2(t)\varphi'_1(t) + \alpha_2^2(t)\varphi'_2(t)}{\alpha_1^2(t) + \alpha_2^2(t)}, \quad (16)$$

$$B_L(t) = \sqrt{\frac{\alpha_1'^2(t) + \alpha_2'^2(t)}{\alpha_1^2(t) + \alpha_2^2(t)} + \frac{\alpha_1^2(t)\alpha_2^2(t)[\varphi_1'(t) - \varphi_2'(t)]^2}{[\alpha_1^2(t) + \alpha_2^2(t)]^2}}. \quad (17)$$

It is also noted that when $\theta(t)$ is less than the predefined threshold ζ , $X(t)$ is regarded as an IMF that satisfying the termination condition. We replace $X(t)$ by $X(t) - m(t)$, and go back to step 1. Finally, the phase signal of pulse-wave are decomposed into multiple IMF components.

2) Pulse-wave signal reconstruction: The pulse-wave generated by the heartbeat not only consists of a single frequency, but also composes of its harmonics with multiple frequencies. Generally, the harmonics of heartbeat are integer multiples of the fundamental frequency, and they play a modulating and enriching role in the shape of the pulse-wave. Therefore, the pulse-wave signal includes the heartbeat and its harmonic frequencies. It is validated that the frequencies of pulse-wave signals consists of the heartbeat frequency and its second and third harmonics [28]. Firstly, the frequency of heartbeat the

with maximum values of all IMFs within the range of 0.8 Hz to 2 Hz are selected. The selected values from all the IMFs are compared, and the IMF mode with largest value is regarded as the fundamental mode of the pulse-wave signal. The fundamental IMF mode is denoted by $IMF_{f_0}(t)$ and the corresponding frequency is noted by f_0 . Then, we select the IMF components in the frequency domain with peaks at $2f_0$ and $3f_0$, respectively, as the second and third harmonics of the heartbeat signal. The corresponding IMF components are respectively denoted by $IMF_{2f_0}(t)$ and $IMF_{3f_0}(t)$. Therefore, the reconstructed pulse-wave signal $S(t)$ can be reconstructed as:

$$S(t) = IMF_{f_0}(t) + IMF_{2f_0}(t) + IMF_{3f_0}(t). \quad (18)$$

At this point, the pulse-wave signal is reconstructed, and the interference and noise has already been filtered out.

E. Designed DSFNN-BP Method

Given the intricate non-linearity in the pulse-wave-blood pressure interplay, constructing a model to encapsulate their relationship poses a formidable challenge. Therefore, this paper proposes a novel dual-stream fusion neural network to complementarily learn more features of pulse-wave, and achieve accurate blood pressure prediction. The structure of the proposed DSFNN-BP method is illustrated in Fig. 2. Considering the different information expressed by features at various scales, the dual-streams (i.e., S-stream and L-stream) adopt different feature learning strategies. Compared to the previous multi-scale fusion model, this paper uses an improved spatial attention mechanism instead of smaller convolutional kernels to recognize the local features of input pulse-wave signal. Specifically, the S-stream employs a spatial attention mechanism to focus on interesting and important areas within the pulse-wave sequence, thereby enhancing the perception of local features. Meanwhile, the L-stream uses larger convolutional kernels to cover a larger receptive field, supplementing the global features and contextual information in the pulse-wave sequence. Applying spatial attention mechanism to the local feature recognition task can provide higher flexibility compared to using fixed-size small convolutional kernels. The spatial attention mechanism allows the model to flexibly focus on the important regions of the input pulse-wave, rather than being limited to the fixed size and shape of small convolutional kernels. After the fusion of the S-stream and L-stream features, the fused features will be input into the multi-head attention module. The aforementioned literature used a combination of LSTM and multi-head attention. Such an operation introduces unnecessary computational overhead without significantly improving the performance. Therefore, this paper only uses the multi-head attention mechanism to extract the temporal representation information of pulse-wave signal. The specific parameters of the proposed network model are listed in Table I.

The pulse-wave signal is input into both the S-stream and L-stream. In the S-stream, to make the d DSFNN-BP identify the local detailed features of the pulse-wave signal, this paper introduces the spatial attention mechanism module in the Convolutional Block Attention Module (CBAM) [29] by replacing

the original two-dimensional convolution kernel with a one-dimensional convolution kernel. This adaptation aligns the spatial attention mechanism with the inherent one-dimensional nature of the pulse-wave signal, allowing it to dynamically assign importance weights to different positions in the signal. Although the basic structure of the spatial attention mechanism remains unchanged, this adaptation optimizes its application to pulse-wave signals, which contributes to more precisely focus on key local features and improves the model's feature extraction performance. The specific calculation of the spatial attention mechanism is as follows:

$$\begin{aligned} \mathbf{M}_S(\mathbf{F}) &= \sigma(f([\text{AvgPool}(\mathbf{F}), \text{MaxPool}(\mathbf{F})])) \\ &= \sigma(f([\mathbf{F}_{avg}^c; \mathbf{F}_{max}^c])). \end{aligned} \quad (19)$$

where σ is the Sigmoid function, and $\mathbf{F}$ is the input feature vector. Initially, we utilize both average-pooling and max-pooling operations on the feature maps, resulting in the derivation of their feature representations. Then, we concatenate the features obtained from both pooling channels and proceed through a convolution operation (denoted as f). Given that the input pulse-wave has only one dimension, we replace the original two-dimensional convolution kernel with a one-dimensional convolution kernel, which kernel size is 3. After passing through the Sigmoid activation operation, the weight coefficients are obtained. Then, we multiply these weight coefficients $\mathbf{M}_S$ by the feature $\mathbf{F}$ to generate new features. This operation helps the network to focus on important detailed features of the pulse-wave.

In the L-stream, this paper employs a convolution-pooling module learning strategy consisting of Conv-BN-MaxPool-ReLU. The convolution-pooling module is utilized to extract global features from pulse-waves. The Batch Normalization (BN) layer effectively prevents overfitting during network training. Stacking Conv-BN-MaxPool-ReLU allows the network's learning gradient to be smoother, and the convergence speed becomes faster as well as the performance becomes better. Finally, the multi-scale feature representations produced by the S-stream and L-stream are concatenated, integrating the multi-feature complementary information of the pulse-wave signal.

TABLE I: Parameters of the DSFNN-BP

Module	Core Parameter	Input Feature Size	Output Feature Size
L-Stream	Kernel1_size = 7 Kernel2_size = 3	[16,1,2500]	[16,32,2498]
S-Stream	Kernel size = 3	[16,1,2500]	[16,32,2498]
Multi-head Attention	Num heads = 8	[16,2498,64]	[16,2498,64]
FC1		[16,2498,64]	[16,128]
FC2		[16,128]	[16,2]

The integrated complementary feature information is input into the multi-head attention mechanism. Multi-head attention utilizes multiple distinct attention heads simultaneously with each head capable of processing multiple parts of the pulse-wave to further enhance the model's ability of capturing different information representation spaces [30]. In the context

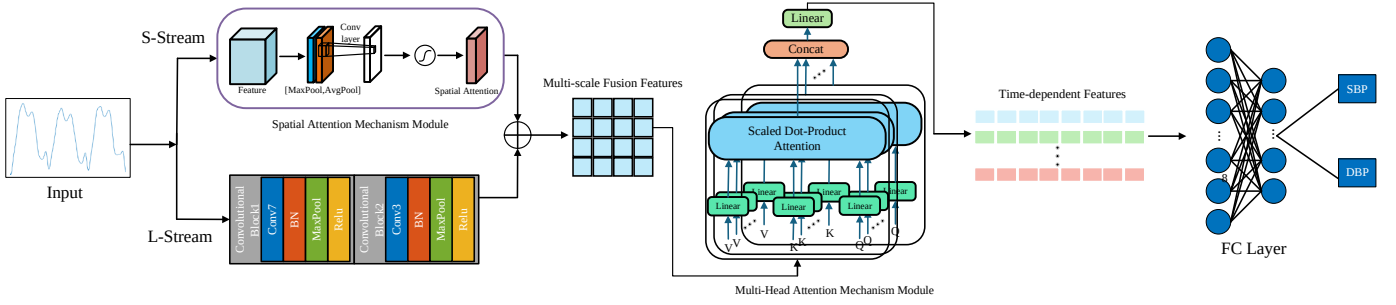


Fig. 2: Schematic of the DSFNN-BP method

of long sequence pulse-wave input vectors, the sequence features at specific positions are crucial for understanding the cardiovascular characteristics implied of the pulse-waves. It is noted that different attention heads focus on dependencies at different time scales. Some attention heads emphasize short-term features of rapid fluctuations and transient changes in the pulse wave, while others capture long-term dependencies of periodic patterns and long-term trends spanning larger time intervals. This enables the integration of information across different time scales, overcoming the limitations of traditional methods in modeling long-term dependencies. The input feature is transformed into queries, keys, and values representations. Then, by calculating the correlation between queries and keys, the attention weights of each position to other positions are obtained. Thus, based on the information of each position in the sequence, the model can calculate the degree of correlation between different positions, focusing on the importance of different times. Typically, multiple independent attention heads compute different attention representations, with each attention head learning different attention weights that can focus on different information in the sequence. The computation of self-attention is as follows:

$$Attention(Q, K, V) = softmax(\frac{QK^T}{\sqrt{d_k}})V \quad (20)$$

where Q , K and V are the query vector, key vector, and value vector, respectively. QK^T is the similarity matrix between the query and key vectors. Dividing by $\sqrt{d_k}$, the magnitude of the inner product can be scaled down to reduce its impact on the gradient. The softmax function normalizes the similarity computed between the query vector at each position and the key vectors at all positions to obtain the attention weights. The final attention representation is obtained by multiplying the value vector matrix with the attention weights.

The computation method of the multi-head attention mechanism is as follow:

$$MH(Q, K, V) = Concat(head_1, \dots, head_h)W^o. \quad (21)$$

where $head_i = Attention(QW_i^Q, KW_i^K, VW_i^V)$. In multi-head attention, W_i^Q , W_i^K and W_i^V are the projection matrices used to project Q , K and V from the input pulse-wave sequence into subspaces Q_i , K_i and V_i , respectively.

Finally, by concatenating the attention representations from all heads and projecting them through another linear trans-

formation W^o , the final multi-head attention output can be obtained. Lastly, the feature vectors generated by the fully connected layer are input into the softmax function for blood pressure estimation.

III. EXPERIMENTAL RESULTS AND ANALYSIS

A. Parameters Configuration and Experimental Setup

This paper utilizes the AWR1642 radar board along with the data acquisition board (DCA1000) to collect the human pulse-wave signals. The Omron Electronic Blood Pressure Monitor is employed to collect the blood pressure ground values, and the measurement accuracy is complied with AAMI standards. The specific radar parameter settings are shown in Table II. Although AWR1642 radar board features a 2-transmitter 4-receiver antenna array, we use only a single-transmitter single-receiver antenna configuration for pulse-wave data collection. The experimental setup is depicted in Fig. 3, the subject lies on the bed in a comfortable position, the radar is positioned about 5 centimeters (without specified) directly above the neck, and the arm is equipped with the blood pressure cuff for comparison. The blood pressure data is recorded using the blood pressure monitor when the radar collects data. It is noted that both the radar and blood pressure monitor perform a data collection duration of 10 seconds.

TABLE II: Radar parameters specifications

Parameters	Specifications	Parameters	Specifications
TX	1	RX	1
Start Frequency	77 GHz	frames	10000
Frequency Slope	70 MHz/s	Chirp per frame	1
Bandwidth	3.99 GHz	Periodicity	1 ms
Idle Time	7 us	ADC Samples	200
Ramp End Time	57 us	Sample Rate	4000 kpsps

This paper collects 40 subjects. The gender ratio of the subjects is 25 males to 15 females, with ages concentrated between 17 and 63 years old, and weights ranging from 45 to 100 kilograms. All subjects have no history of cardiovascular diseases. To increase the diversity of the experimental samples, the subjects consume caffeinated beverages to induce changes in blood pressure. We collected data at different time periods. Test participants need a 5-minute rest interval between each test. The duration of each piece of data is 10 seconds, containing more than 10 pulse cycles. By taking 3 consecutive pulse cycles as a data group, a sample data set of 2500 pieces

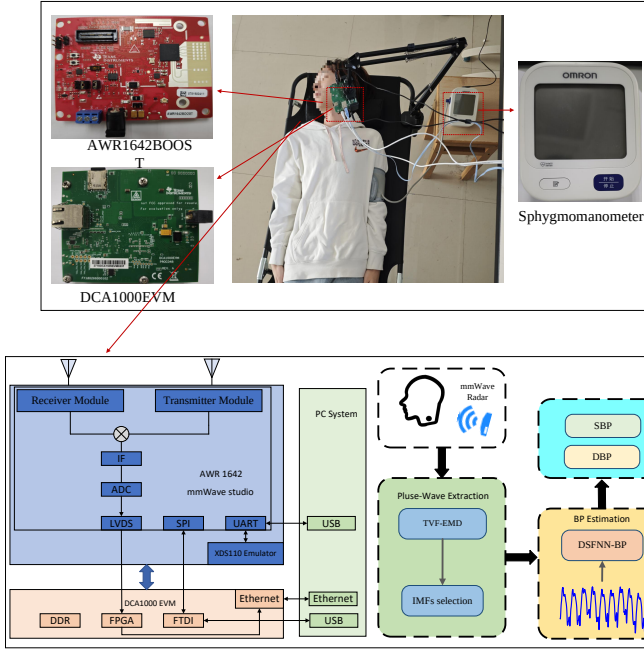


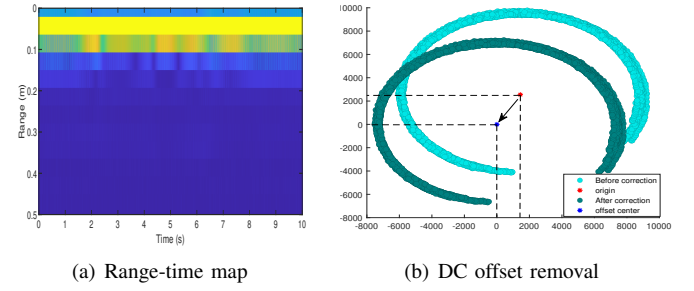
Fig. 3: Experimental scenarios and equipment

is obtained. The test set and the training set are randomly assigned in a ratio of 7:3, and input to a designed DSFNN-BP network.

B. Results and Analysis

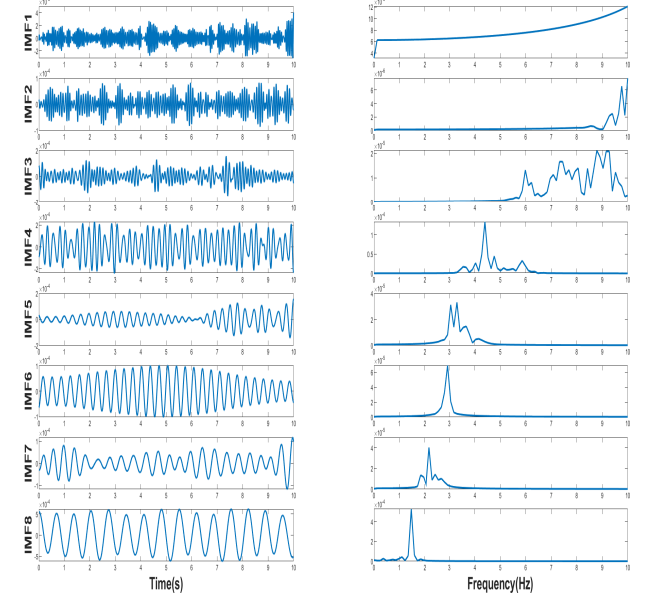
Detailed results of the proposed method: Due to the use of a single-transmit-single-receive antenna, only one data channel is available. FFT is applied to construct RTM as shown in Fig. 4(a). We select the brightest region in the RTM image as the location of the pulse signal, and extract the corresponding I/Q signals. After removing the DC bias (see Fig. 4 (b)), we obtain the phase information of the pulse signal. Employing the TVF-EMD algorithm, we decompose the phase signal into a series of IMF components, as depicted in Figure 4(c). The figure shows that the maximum peak in the spectrum is 1.46 Hz. We select the corresponding IMF8 as the main IMF to reconstruct the pulse-wave. The spectral peak of its second harmonic is 2.92 Hz, and the spectral peak of its third harmonic is 4.38 Hz. We select IMF6 and IMF4, respectively, and then sum these three IMFs to yield the reconstructed pulse-wave signal. The denoising reconstruction results are shown in Figs 4(d) and (e). Compared to the original phase signal, the reconstructed pulse-wave signal obtained by TVF-EMD allows for precisely identifying the detailed features of the pulse-wave, not only low-frequency and high-frequency noise, but also noise within the frequency domain range of the pulse-wave has been removed, which provides favorable conditions for the subsequent blood pressure monitoring.

According to IEEE standards, we used the Bland-Altman plots to reflect the experimental results. As shown in Figs. 5 and 6, the central dashed line depicts the average discrepancy between the observed and predicted values, while the superior and inferior lines frame the 95% confidence interval boundaries, which are 1.96 standard deviations above and below

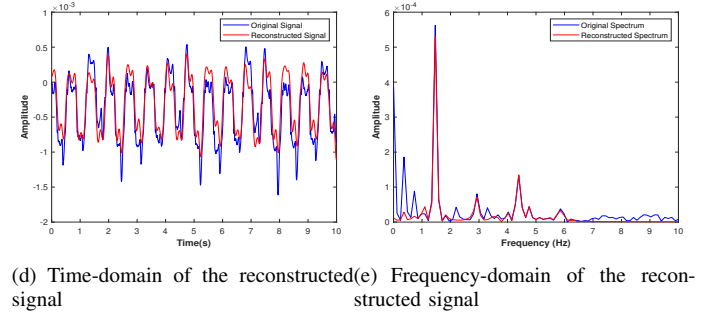


(a) Range-time map

(b) DC offset removal



(c) TVF-EMD decomposition of pulse-wave signal



(d) Time-domain of the reconstructed signal (e) Frequency-domain of the reconstructed signal

Fig. 4: Signal decomposition and reconstruction results

this mean difference, respectively. The majority of data points fall within the 95% consistency interval, indicating that the agreement between the predicted and actual results is good. The Mean Error (ME) and Standard Deviation (SD) for SBP and DBP obtained in our experiments are (-0.49, 5.19) and (-0.32, 5.11), respectively.

C. Hyperparameter Analysis of DSFNN-BP

In the training process of the DSFNN-BP network, the learning rate and batch size play a crucial role in model perfor-

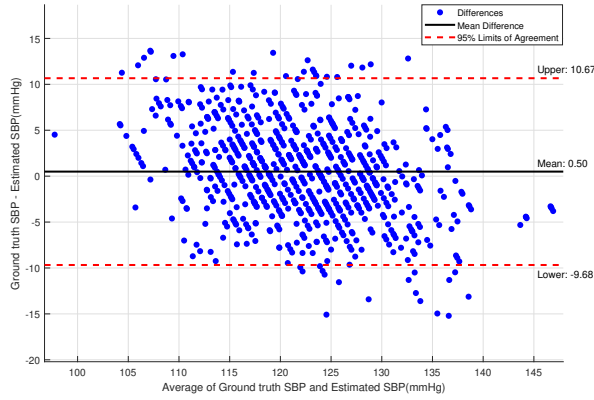


Fig. 5: Bland-Altman diagram of estimation results in SBP

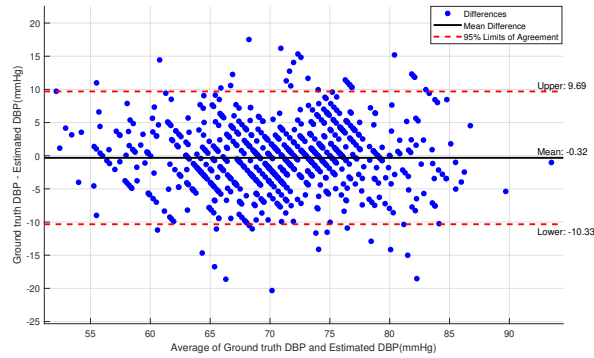


Fig. 6: Bland-Altman diagram of estimation results in DBP

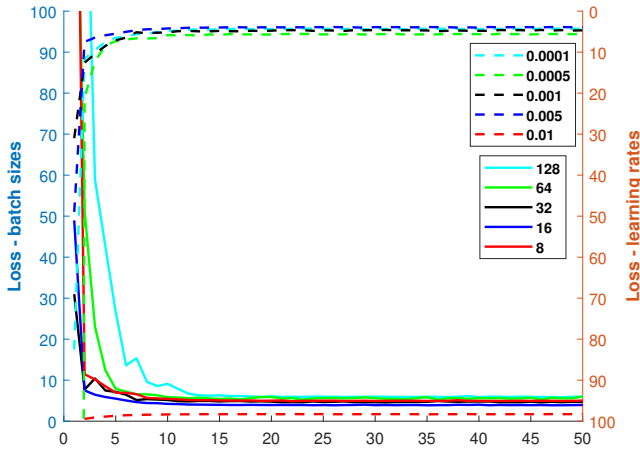


Fig. 7: Effect of learning rate and batch size on loss function

mance and convergence. The initial learning rate significantly influences the network's convergence behavior. Experiments with learning rates of 0.0001, 0.0005, 0.001, 0.005, and 0.01 reveal that excessively high learning rates, such as 0.01, prevent convergence, while excessively low learning rates, such as 0.0001, results in slow convergence and long training time. As shown in Fig. 7, the learning rate of 0.0005 achieves the lowest loss with optimal convergence, and is therefore selected for subsequent experiments. Similarly, to investigate the impact of batch size on blood pressure prediction performance, the network training parameters are kept consistent while setting the batch sizes to 8, 16, 32, 64, and 128. The results indicate that a smaller batch sizes leads to faster convergence but higher fluctuations, while a larger batch sizes slows down the convergence process. A batch size of 16 achieves the best trade-off with stable loss curves and fast convergence. Therefore, in the subsequent experiments, we adopt a batch size of 16.

D. Comparison with the State-of-the-Art methods

(1) Denoising Effect Comparison: To validate the effectiveness of the proposed method in this paper, we analyze the impact of different denoising algorithms on the denoising results, and the results are illustrated in Fig. 8. As shown in Fig. 8, the pulse wave obtained by the comparative denoising algorithms exhibits a clear morphology, but noise cannot be completely removed. As shown in Fig. 8(c) and (d), the phase-wave signals decomposed by EMD and ICEMMDAN still retain some noise, even though part of the noise has been eliminated compared to the original phase signals. This is attributed to the mode mixing of the decomposed signal components, specifically the mixing of the third harmonic with high-frequency noise. By utilizing TVF-EMD decomposition and reconstruction, the resulting signal retains the characteristics of the pulse wave in all aspects, demonstrating the validity of the denoising method presented in this paper.

(2) Training Comparison: The denoised pulse wave sequences are input into different neural networks for blood pressure prediction. Fig. 9 shows a comparison of the training processes in terms of loss value. The loss function used to train the model is L1 Loss (known as MAE), which is commonly used in regression tasks. It can be observed that DSFNN-BP converges faster compared to other comparison methods. The DSFNN-BP model employs a dual-stream structure to extract large-scale and small-scale features in parallel and combines with an attention mechanism to adaptively fuse important features, thereby avoiding redundant computations. Such an operation significantly enhances the feature extraction efficiency and training speed, enabling the model to quickly focus on key features for blood pressure prediction. As a result, the proposed DSFNN-BP achieves a faster error reduction in the training process.

(3) Comparison of Recognition Methods: We discussed the impact of different neural networks on the blood pressure monitoring performance. The neural networks we have compared are the prevailing methods currently utilized in the field of blood pressure prediction. We used CNN, CNN + LSTM,

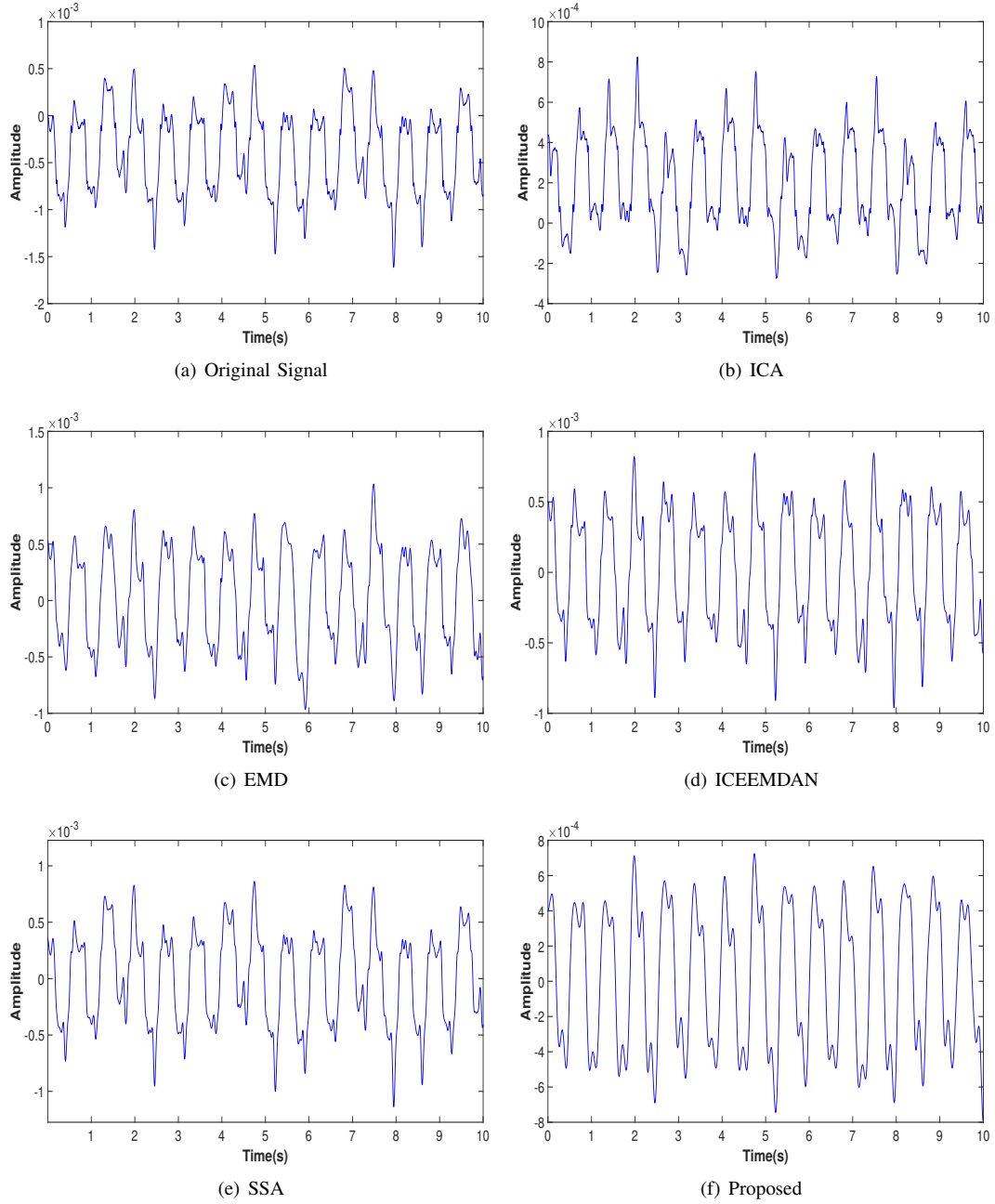


Fig. 8: Comparison of the denoising methods

attention-based RNN, and TRCCBP as comparative methods. (1) The CNN method use CNN model to extract local spatio-temporal features from the pulse wave with convolutional layers to capture local patterns in blood pressure changes. (2) CNN + LSTM combines CNN (for local feature extraction) with LSTM to model temporal dependencies, allowing the model to process both spatial and temporal features of the pulse wave for blood pressure prediction. (3) Attention-based RNN incorporates an attention mechanism into the RNN to focus on critical parts of the pulse wave, dynamically selecting key features for blood pressure prediction. (4) TRCCBP uses the convolutional kernels of different sizes, a GRU module, and an attention mechanism to extract multi-scale spatio-

temporal features from the pulse wave, capturing key information for blood pressure prediction. Since the prediction results are heavily influenced by the dataset, all the neural networks we compared utilized the same dataset for fairness and consistency. We use the MD $\pm$ SD proposed by ANSI/AAMI SP, and the MAE introduced by IEEE standards, and the Mean Absolute Percentage Error (MAPE) specified by the British Hypertension Society (BHS) protocol as the evaluation indicators. It is noted that the MAPE is a percentage that represents the percentage of average estimation error. According to the data presented in Table III, our proposed model has achieved the best results across all evaluated indicators, outperforming other methods in comparison. The MAE for SBP and DBP

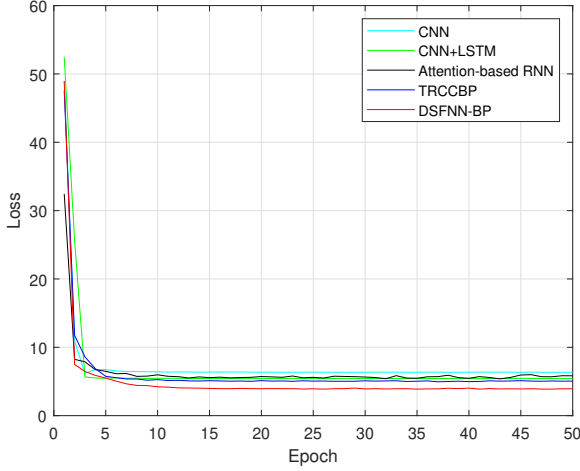


Fig. 9: Recognition loss comparison

are 4.16 mmHg and 3.87 mmHg, respectively, and the MAPE for SBP and DBP are 5.23 and 4.91, respectively. According to the new grading system based on MAE precision levels by IEEE, the results obtained based on proposed method can reach Class A ($MAE \leq 5$ mmHg). Compared with other deep learning methods, the proposed model can adaptively capture the detailed features and global features of the pulse-waves, and fully extract the information of the pulse-waves for blood pressure prediction. In all, the results indicate that the our DSFNN-BP method has achieved a more favorable experimental results compared to other methods.

(4) Model size and running time comparison: Real-time inference time is a crucial aspect in practical applications, especially for real-time blood pressure measurements. We compared the running time and the model size of different models, and the results are shown in Table IV. The model size of our DSFNN-BP is 79.66 MB, indicating that the method is suitable for running on resource-constrained devices. The model's single inference time is 136 ms, which meets the requirements for real-time inference. Although the average running time of the proposed method is slightly longer than that of existing methods, it delivers superior blood pressure monitoring performance and remains suitable for real-time applications.

TABLE III: Performance comparison of different networks for SBP and DBP

Methods	SBP			DBP		
	MD(SD)	MAE	MAPE	MD(SD)	MAE	MAPE
CNN	1.70(8.51)	6.70	8.25	-1.46(8.01)	6.51	8.13
CNN+LSTM	1.27(6.92)	5.21	6.67	-0.92(7.29)	5.23	6.09
Attention-based RNN	1.53(7.14)	5.99	7.15	-1.30(6.10)	5.64	6.12
TRCCBP [18]	0.94(6.28)	4.95	5.89	-0.62(6.46)	4.53	5.77
Proposed	-0.49(5.19)	4.16	5.23	-0.32(5.11)	3.87	4.91

E. Effect of Distance

In this subsection, we test the impact of different distances (5 cm, 10 cm, 15 cm, 20 cm) from the radar to the human

TABLE IV: Model size and running time comparison of different networks

Networks	Model Size (MB)	Single Inference Time (ms)
CNN	69.51	166
CNN+LSTM	157.26	262
Attention-based RNN	57.71	83
TRCCBP [18]	94.84	149
Proposed	79.66	136

body neck on the monitoring performance of blood pressure. The error results are shown in Fig.10, which indicate that MAEs of both SBP and DBP is positively correlated with changes in distance. This may be attributed to signal distortion caused by environmental noise and multipath effects, making it challenging to obtain the features of pulse-wave signals. Specially, when the distance is within 5 cm, MAEs of both SBP and DBP satisfy the AAMI standard ($MAE \leq 5$ mmHg). However, when the distance is larger than 15 cm, only the MAE of SBP meets the requirements of AAMI. When the distance exceeds 30 cm, there is a significant decrease in the accuracy of blood pressure prediction, it seems to be challenging to extract useful pulse-wave signals for blood pressure monitoring. This is because as the distance increases, the received signal strength weakens due to the increase of attenuation, and environmental interference and noise. As a result, the weaker signal makes the detection method more difficult to distinguish the target signal from noisy signal, leading to a reduced detection accuracy.

F. Effect of Angle

In this subsection, we test the different angles (-60° , -30° , 0° , 30° , 60°) from the radar to the human body neck on the monitoring performance of blood pressure. The results shown in Fig.11 indicate that when the angle is 0° , we get the best MAE performance. When the angle are smaller than $\pm 30^\circ$, the MAE performance meets the AAMI requirement. Unfortunately, when the radar angle reaches $\pm 60^\circ$, the waveform of the pulse-wave experiences severe distortion and damage, making it difficult to extract accurate pulse-wave signals. As the angle increases, the radial motion component between the radar and the human target decreases. Since the radar can only capture the motion changes along its radial direction, a larger angle reduces the amplitude of these motion changes. This reduction increases the relative interferences, making it difficult to accurately extract blood pressure signals. It is noteworthy that the positive and negative values of the angle seem to have a certain impact on the results, which may be related to the physiological structure of the human head and neck.

IV. CONCLUSION

This paper proposed a contactless blood pressure monitoring method based on millimeter-wave FMCW radar. We utilized the millimeter-wave radar to capture minute skin vibration signals from the human neck, and decomposed the phase signal into several IMF components by TVF-EMD, we reconstructed the pulse-wave signal by selecting the heartbeat signal, and its

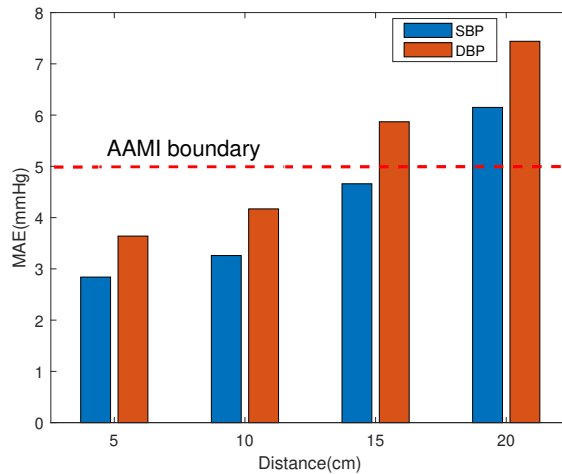


Fig. 10: MAE at Different Distances

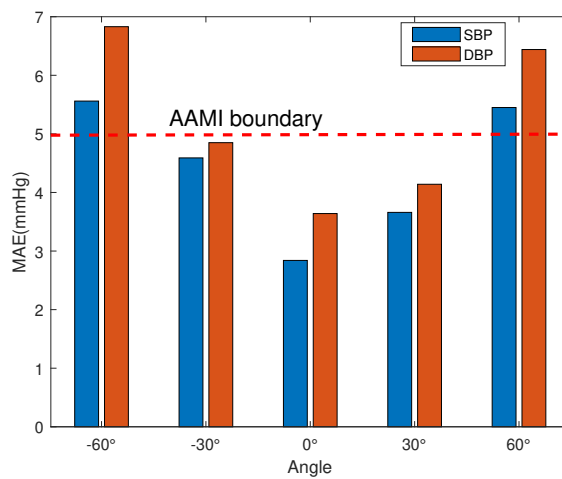


Fig. 11: MAE at Different Angles

second as well as third harmonics. Finally, we proposed a novel DSFNN-BP method for extracting pulse-wave features to achieve blood pressure estimation. The results of experimental demonstrated that our proposed method complied with the standards proposed by the AAMI, and had the best monitoring performance compared to the state-of-the-art methods. In the future, we plan to address blood pressure detection issues in various scenarios.

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